Semantic Correspondence with Visual Foundation Models

Mario De Paola
s334090@studenti.polito.it

Tommaso Centonze
s338411@studenti.polito.it

Giuseppe Latino
s339356@studenti.polito.it

Lucio Fuoco
348016@studenti.polito.it

Abstract

In this work, we address the problem of semantic correspondence using visual foundation models. We evaluate the zero-shot performance of DINOv2, DINOv3, and SAM on the SPair-71k benchmark, analyzing their ability to capture semantic-level correspondences without task-specific training. We then investigate light fine-tuning strategies and show that selectively adapting a limited number of layers leads to an improvement of PCK over frozen backbones. Furthermore, we introduce a window soft-argmax that improves spatial precision and yields an additional gain. Finally, we analyze the generalization behavior of the proposed approach across different datasets, highlighting the robustness of foundation-model-based representations for semantic correspondence.
Our code is publicly available here

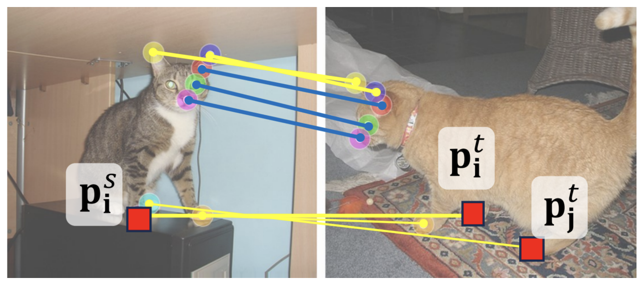

Figure 1. Annotations of geometry aware semantic correspondence (yellow) and standard semantic correspondence (blue)

1. Introduction

Semantic correspondence aims to establish point-to-point matches between images depicting different instances of the same object category. Unlike classical feature matching, the goal is to align semantically corresponding parts despite large variations in appearance, pose, scale, and background. This problem is fundamental to object-level alignment and fine-grained visual understanding, but local visual similarity often fails under intra-class variation and viewpoint changes. Large-scale benchmarks such as SPair-71k enable systematic evaluation of correspondence methods under these challenging conditions [6]. Recent progress in large-scale pre-training has produced visual foundation models that learn rich representations without task-specific supervision. When used as dense feature extractors, such models provide a promising prior for semantic correspondence. In particular, self-supervised Vision Transformers yield robust patch-level features with strong transfer properties [7, 8], while diffusion-based generative models have been shown to produce meaningful dense descriptors that support correspondence in a zero-shot setting [9]. However, recent studies indicate that even strong pretrained representations are not sufficient in all cases. Geometric ambiguities such as symmetric parts and mirrored configurations can lead to systematic correspondence errors, highlighting limitations in purely appearance-based matching [12]. In this work, we study semantic correspondence using pretrained foundation models as dense feature extractors. We estimate correspondences via cosine similarity between dense features and evaluate performance using PCK on SPair-71k [6]. We analyze the impact of lightweight fine-tuning by adapting only a small subset of backbone layers and improve localization accuracy using a window soft-argmax decoding strategy. Finally, we assess cross-dataset transfer on PF-Pascal, PF-Willow, and AP-10K [2, 10] to characterize how backbone choice and light adaptation affect generalization. In addition to self-supervised and generative models, we include SAM [3] as a segmentation-oriented foundation baseline, exploring whether representations encoding object boundaries and structure can support correspondence in cluttered scenes.

2. Related Work

2.1. Self-supervised Vision Transformers

Self-supervised Vision Transformers learn general-purpose visual representations that exhibit strong transfer across downstream tasks. Early work demonstrated emergent object-centric and part-aware structure in ViT features trained without supervision [1]. These models are well suited for correspondence when repurposed as dense feature extractors.

2.2. Diffusion-based dense descriptors

Diffusion models, originally developed for image generation, have recently been repurposed as dense feature extractors for correspondence. Tang et al. showed that intermediate diffusion features encode spatially consistent descriptors that enable strong zero-shot correspondence performance [9]. Beyond their standalone performance, diffusion features often capture complementary cues compared to self-supervised ViTs. Combining diffusion and ViT-based descriptors has been shown to significantly improve correspondence accuracy on challenging benchmarks such as SPair-71k [11].

2.3. Geometry-aware correspondence

Despite the strength of foundation representations, correspondence remains difficult in the presence of geometric ambiguities. Recent work highlights systematic failures on symmetric parts and mirrored configurations, motivating geometry-aware analyses and disambiguation strategies [12]. Our work complements these efforts by focusing on practical design choices, including backbone selection, lightweight adaptation depth and decoding strategies, that influence robustness and generalization.

2.4. Segmentation-oriented foundation models

Segment Anything (SAM) provides promptable segmentation across diverse categories and yields representations that encode object boundaries and region-level structure [3]. Although not trained for correspondence, such segmentation-oriented features may support correspondence pipelines by emphasizing object structure, particularly in cluttered scenes. We include SAM as an exploratory foundation baseline within a unified correspondence framework.

3. Methodology

We study semantic correspondence using a simple and modular pipeline based on dense features extracted from pre-trained foundation backbones. Given a source image I_s and a target image I_t, we (i) extract dense feature maps, (ii) compute similarity-based correlation maps, and (iii) decode correspondences from correlation maps. We evaluate both the zero-shot setting and a lightweight fine-tuning regime where only the last k layers of the backbone are adapted.

3.1. Problem Formulation

Given two images I_s (source) and I_t (target) depicting different instances of the same object category, semantic correspondence aims to estimate a mapping from points in I_s to their corresponding locations in I_t. For a query point $\mathbf{p}_s \in \mathbb{R}^2$ in the source image, we predict its correspondence $\mathbf{p}_t \in \mathbb{R}^2$ in the target image, despite variations in appearance, pose, scale, and background.

Although ground-truth annotations in semantic correspondence benchmarks are sparse, dense feature matching approaches [11] enable predictions at any image location.

3.2. Dense Feature Extraction

We use pretrained backbones as dense feature extractors. Given an image I, a backbone network $f_\theta(\cdot)$ produces a feature map

$$\mathbf{F} = f_\theta(I) \in \mathbb{R}^{C \times H \times W} \tag{1}$$

where C is the feature dimensionality and $H \times W$ is the spatial resolution. Due to differences in patch size across backbones, we select the input image resolution such that all models produce feature maps with identical spatial resolution, enabling a fair comparison.

3.3. DINOv2 and DINOv3 backbones.

We employ DINOv2 and DINOv3 Vision Transformer backbones as dense feature extractors. For both models, we adopt a unified extraction strategy: the input image is processed through the transformer encoder, and features are retrieved from the output of the last transformer block. We retain only the spatial patch tokens, discarding the global `[CLS]` token to isolate local semantic information. In the case of DINOv3, which introduces a small set of auxiliary register tokens to decouple global context, we additionally remove these four register tokens prior to reshaping. The remaining patch sequence (B, N, C) is then reshaped into a 2D feature grid $(B, C, H/p, W/p)$, where p denotes the patch size of the Vision Transformer, preserving the spatial alignment required for dense correspondence

3.4. Segment Anything (SAM) backbone.

To leverage the strong semantic priors of the Segment Anything Model (SAM) [3], we repurpose its ViT-based image encoder as a dense feature extractor. Unlike standard ViTs, SAM is trained at a fixed input resolution of 1024×1024 and with model-specific pixel statistics. To support arbitrary input sizes and aspect ratios, we dynamically adapt SAM's positional embeddings by resizing the pretrained positional weights via bilinear interpolation to match the spatial resolution of the current token grid. Finally, we apply SAM's original input normalization during preprocessing to ensure that features remain consistent with the model's training distribution.

3.5. Similarity-Based Matching

To enable similarity-based matching, we ℓ_2-normalize features along the channel dimension:

$$\tilde{\mathbf{F}}(\mathbf{x}) = \frac{\mathbf{F}(\mathbf{x})}{\|\mathbf{F}(\mathbf{x})\|_2 + \epsilon} \quad (2)$$

with a small ϵ for numerical stability. Given normalized source and target feature maps $\tilde{\mathbf{F}}_s$ and $\tilde{\mathbf{F}}_t$, we compute correspondences using cosine similarity. For a source location $\mathbf{x}$, the similarity to a target location $\mathbf{y}$ is

$$\mathrm{sim}(\mathbf{x}, \mathbf{y}) = \tilde{\mathbf{F}}_s(\mathbf{x})\, \tilde{\mathbf{F}}_t(\mathbf{y})^\top \quad (3)$$

This produces a correlation map over the target image for each source location. In practice, for keypoint matching we only compute correlation maps for annotated (or queried) source points.

3.6. Correspondence Decoding

We evaluate two strategies to decode correspondences from the correlation maps: a standard *argmax* and a *window soft-argmax*.

3.6.1. Argmax

Argmax consists of selecting the target location with the highest cosine similarity score. For a source point $\mathbf{x}$, the predicted target $\mathbf{p}_t$ is:

$$\mathbf{p}_t = \arg\max_{\mathbf{y}} \mathrm{sim}(\mathbf{x}, \mathbf{y}). \quad (4)$$

This method is efficient, but restricted to the discrete spatial resolution of the feature grid and can be sensitive to noise in repetitive textures.

3.6.2. Window Soft-Argmax

To address the limitations of the hard argmax and achieve sub-pixel precision, we employ a window soft-argmax strategy [12]. We first identify the coarse location $\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \mathrm{sim}(\mathbf{x}, \mathbf{y})$ and define a local window $\mathcal{W}$ centered at $\hat{\mathbf{y}}$. We then compute the expected coordinate within this neighborhood:

$$\mathbf{p}_t = \sum_{\mathbf{y} \in \mathcal{W}} \mathbf{y} \cdot \frac{\exp(\mathrm{sim}(\mathbf{x}, \mathbf{y})/\tau)}{\sum_{\mathbf{y}' \in \mathcal{W}} \exp(\mathrm{sim}(\mathbf{x}, \mathbf{y}')/\tau)}, \quad (5)$$

where τ is a temperature parameter controlling the sharpness of the distribution. This refinement provides smoother predictions and enforces local consistency, mitigating the effect of distant outliers.

3.7. Lightweight Fine-Tuning

While pretrained backbones provide strong representations in a zero-shot setting, they are not explicitly optimized for semantic correspondence. We explore a lightweight adaptation strategy by freezing most backbone layers and fine-tuning only the last k, with $k \in \{1, 2, 4\}$. This preserves general-purpose representations while allowing limited task-specific specialization.

3.7.1. Fine-tuning objective

We employ a correlation-based supervision objective operating on dense feature maps. Given annotated correspondences $\{(\mathbf{p}_s^i, \mathbf{p}_t^i)\}_{i=1}^N$, we sample the source descriptor at $\mathbf{p}_s^i$ from $\tilde{\mathbf{F}}_s$ using bilinear interpolation, and compute a dense correlation score with all target locations:

$$s_i(\mathbf{y}) = \mathbf{f}_s^{i\top} \tilde{\mathbf{F}}_t(\mathbf{y}) \quad (6)$$

where $\mathbf{f}_s^i \in \mathbb{R}^C$ is the sampled (normalized) source feature. We convert scores into a probability map via a softmax with temperature β:

$$P_i(\mathbf{y}) = \frac{\exp(s_i(\mathbf{y})/\beta)}{\sum_{\mathbf{y}'} \exp(s_i(\mathbf{y}')/\beta)} \quad (7)$$

Instead of one-hot supervision, we apply a Gaussian-smoothed cross-entropy loss [4] in a local neighborhood around the ground-truth correspondence. Let $\mathcal{W}_K(\mathbf{p}_t^i)$ denote a $K \times K$ window centered at $\mathbf{p}_t^i$, and let G_K be a normalized Gaussian kernel over the window offsets. The per-keypoint loss is

$$\mathcal{L}_i = -\sum_{\mathbf{y} \in \mathcal{W}_K(\mathbf{p}_t^i)} G_K(\mathbf{y} - \mathbf{p}_t^i) \log P_i(\mathbf{y}) \quad (8)$$

The final loss averages $\mathcal{L}_i$ over annotated keypoints and valid samples in the batch. This provides smooth spatial supervision and mitigates discretization and annotation noise.

3.8. Pipeline Summary

Our correspondence pipeline consists of: (1) extracting dense features from a pretrained backbone, (2) computing cosine-similarity correlation maps between source and target features, (3) decoding correspondences via windowed soft-argmax, and (4) optionally adapting the backbone via lightweight fine-tuning of the last k layers using the correlation-based objective above.

4. Experiment

We first describe our evaluation protocol and experimental setup, designed to place all considered backbones under identical conditions for a fair comparison. Within this unified setting, we analyze both frozen and lightly fine-tuned variants of each model, and we study the effect of different correspondence inference operators. We report results on SPair-71k as our main benchmark and additionally evaluate on PF-Pascal, Willow, and AP-10K to assess robustness across datasets and object categories. Overall, we find that light fine-tuning consistently improves performance over frozen pretrained features. Moreover, replacing argmax with a window soft-argmax yields further gains in zero-shot semantic correspondence, indicating that smoother, locally constrained localization can improve matching accuracy.

In this section, we present a comprehensive evaluation of foundation models for semantic correspondence. To ensure a fair comparison, we establish a unified protocol that analyzes both frozen and lightly fine-tuned variants of DINOv2, DINOv3, and SAM under identical conditions. Our experimental analysis primarily focuses on SPair-71k to benchmark performance gains driven by lightweight adaptation and windowed soft-argmax decoding. Finally, we assess the generalization capabilities of our optimized models through cross-dataset evaluation on PF-Pascal, PF-Willow, and AP-10K.

4.1. Evaluation Details

Evaluation Protocol We follow the standard protocol from DIFT[9] to evaluate semantic correspondence using the Percentage of Correct Keypoints (PCK) metric. Given an image pair (I_A, I_B) and its associated correspondence set $\mathcal{X} = \{(\mathbf{x}_q^A, \mathbf{x}_q^B)\}_{q=1}^n$, for each $\mathbf{x}_q^A = (x_q^A, y_q^A)$ we find its predicted correspondence $\hat{\mathbf{x}}_q^B$ and compute PCK for the image pair as

$$\mathrm{PCK}(I_A, I_B) = \frac{1}{n}\sum_{q=1}^{n} \mathbb{I}\big(\|\hat{\mathbf{x}}_q^B - \mathbf{x}_q^B\| \le \alpha\,\theta\big), \qquad (9)$$

where θ is the base threshold, $\alpha < 1$ is the normalized tolerance, and $\mathbb{I}(\cdot)$ is the binary indicator function. For PF-Pascal, we set $\theta_{\mathrm{img}} = \max(h_{\mathrm{img}}, w_{\mathrm{img}})$. For PF-Willow, we use $\theta_{\mathrm{kps}} = \max\big(\max_q(x_q^B) - \min_q(x_q^B),\ \max_q(y_q^B) - \min_q(y_q^B)\big)$. For SPair-71k, we set $\theta_{\mathrm{bbox}} = \max(h_{\mathrm{bbox}}, w_{\mathrm{bbox}})$, where h_{bbox} and w_{bbox} are the height and width of the bounding box. For AP10K, we set $\theta_{bbox} = \max(h_{bbox}, w_{bbox})$, where h_{bbox} and w_{bbox} are the height and width of the object bounding box. These choices follow standard conventions used in prior work. We report results under two complementary aggregation schemes.

PCK per image computes the average PCK independently for each image pair and then averages across the dataset, thus giving equal weight to all image pairs.

PCK per point instead aggregates correctness across all keypoints in the dataset, assigning higher weight to image pairs with more annotations.

4.2. Benchmark Datasets

We evaluate our approach on four benchmark datasets that capture complementary challenges in semantic correspondence:

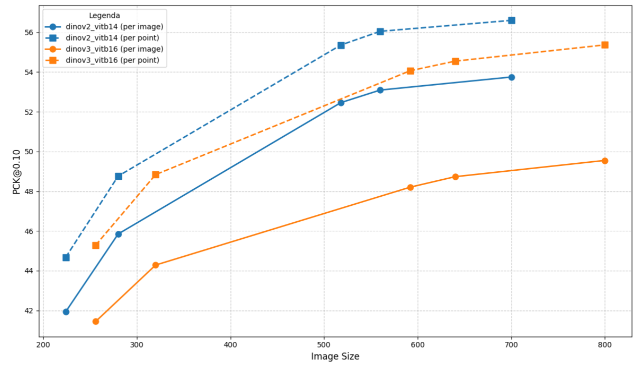

Figure 2. Comparison between DINOv2 and DINOv3 at varying image sizes. PCK@0.10 is reported both per image and per point.

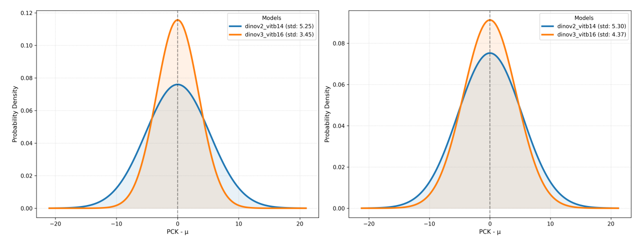

Figure 3. Zero-mean Gaussian fits of PCK@0.10 deviations across image resolutions are shown for DINOv2 and DINOv3, computed per image (left) and per point (right), where a smaller standard deviation indicates higher robustness to resolution changes.

(a) SPair-71k [6] is our primary benchmark. It contains 70,958 image pairs with significant intra-class variations in viewpoint, scale, and appearance. Unlike smaller datasets, it provides a rigorous test for dense semantic matching under large geometric deformations.

(b) PF-Pascal [2] consists of 1,351 image pairs from 20 PASCAL categories. It is a widely recognized benchmark that allows for a direct comparison with state-of-the-art methods under a standardized evaluation protocol.

(c) PF-Willow [2] is a classic dataset featuring 900 image pairs across 5 object classes. With its sparse landmark annotations, it is primarily used to assess the precision of point-level matching in more constrained scenarios.

(d) AP-10K [10] is a large-scale, in-the-wild animal pose dataset. We leverage its diverse biological taxonomies to evaluate the cross-domain generalization of our model beyond the standard PASCAL-based categories.

4.3. Impact of Image Size on Performance

We analyze the impact of the input image resolution on semantic correspondence performance by comparing DINOv2 and DINOv3 backbones across multiple image sizes. Figure 2 reports the PCK@0.10 scores computed both per image and per point. For both models, performance consistently improves as the image resolution increases, highlighting the importance of high-resolution features for dense correspondence. Tests conducted on SAM further confirm this trend, showing superior performance at 1024×1024 compared to 592×592. The comparison reveals distinct behaviors between the two architectures. At lower resolutions, the performance gap is minimal, with DINOv3 remaining highly competitive and even slightly outperforming DINOv2 in per-point scores. However, DINOv2 scales significantly better: the gap in its favor widens as the resolution increases, allowing it to achieve superior peak performance in high-resolution regimes.In terms of stability, defined as robustness to resolution changes, DINOv3 proves to be the more stable model. This is quantitatively supported by the lower standard deviation of its PCK scores across the evaluated resolutions compared to DINOv2. As show in Figure 3, a lower standard deviation indicates that DINOv3's performance fluctuates less as the input size varies, suggesting that while DINOv2 is preferable for tasks allowing high-resolution input, DINOv3 provides more consistent representations across varying scales. Despite the clear benefits of higher input resolutions, for correspondence estimation we resize all input images to resolutions that yield feature maps with identical spatial dimensions (37×37) across backbones, ensuring a fair architectural comparison. Under this constraint, DINOv2 is evaluated at a resolution of 518×518, while DINOv3 is evaluated at 592×592. For consistency, the same 592×592 resolution is also adopted for SAM. This design choice guarantees alignment of the feature grids across models while balancing computational cost and inference time.

4.4. Backbone Comparison Across PCK Thresholds

Table 1 summarizes the performance of different backbone families under a unified evaluation protocol, reporting PCK both per image and per point. Within each model family, the larger ViT-B variant consistently outperforms ViT-S across thresholds, indicating that higher-capacity encoders provide more discriminative features for semantic correspondence. Comparing across backbones, DINOv2 achieves the best overall performance, with a consistent margin over DINOv3 under both aggregation schemes. In contrast, SAM lags behind DINO-based representations by a large margin, suggesting that features learned for promptable segmentation transfer less effectively to correspondence when used as generic descriptors. Finally, we note that relative trends are stable between the per-image and per-point metrics, supporting the robustness of the observed ranking across different aggregation schemes.

4.5. Fine-tuning Details

We fine-tune pretrained backbone models, including DINOv2 [7], DINOv3 [8], and SAM [3], on the training split of SPair-71k [6]. Fine-tuning is performed for 5 epochs using the AdamW optimizer [5] with a learning rate of 1×10^{-5} and weight decay of 1×10^{-2} . For each backbone, we unfreeze the last $X \in \{1, 2, 4\}$ transformer layers, while keeping the remaining frozen. All models are trained using the correspondence loss described in Sec. 3. Unless otherwise specified, we use a fixed softmax temperature $\tau = 0.04$ during training, which is kept constant across all experiments. We use a batch size of 32 for DINOv2 and DINOv3 across all fine-tuning configurations, as well as for SAM when unfreezing one or two layers. Due to higher computational costs, SAM fine-tuning with $X = 4$ is performed with a reduced batch size of 16. All other training settings are kept identical across backbones to ensure a fair comparison. During fine-tuning, we evaluate performance at the end of each epoch on the validation split using PCK@0.10 per keypoint. The final checkpoint is selected based on the best validation PCK score rather than the final training loss, since lower loss does not necessarily correlate with improved correspondence accuracy.

4.6. Frozen vs Light Fine-tuning

Table 2 and Table 3 report the impact of light fine-tuning on DINO-based backbones. A substantial performance gap is observed between frozen models (L0) and lightly fine-tuned variants, confirming that task-specific adaptation is crucial for semantic correspondence. Across all backbones, unfreezing only the last layer (L1) already yields a large performance jump relative to the frozen setting, accounting for most of the improvement observed with deeper adaptation.

In particular, for DINOv2, fine-tuning the last two transformer layers (L2) yields the best performance in terms of PCK per image, while unfreezing four layers (L4) provides slightly higher PCK per point. Overall, L2 offers the most favorable trade-off between performance and the number of trainable parameters, suggesting that limited adaptation is sufficient to align pretrained representations with the correspondence task. Further increasing the number of trainable layers does not consistently improve performance and may lead to minor degradation, indicating potential overfitting or disruption of pretrained features.

For DINOv3, we select L2 as our default fine-tuning configuration since it achieves the best results under both aggregation schemes (PCK per image and per point) while remaining close in complexity to L1.

SAM follows the same trend of a strong gain already at L1, but it continues to benefit more noticeably from unfreezing additional layers (Table 4). While frozen SAM features perform poorly for semantic correspondence, progressively fine-tuning a larger portion of the backbone leads to consistent improvements. The best performance is achieved by unfreezing four layers, indicating that SAM requires

Backbone	Var.	@0.05	@0.10	@0.15	@0.20
DINOv2	ViT-S	33.61	50.12	59.57	66.14
DINOv2	ViT-B	**35.88**	**52.46**	**64.82**	**68.03**
DINOv3	ViT-S	30.02	46.06	55.11	61.57
DINOv3	ViT-B	**32.73**	**48.20**	**56.51**	**62.93**
SAM	ViT-B	13.98	27.53	27.78	33.84

(a) PCK per image.

Backbone	Var.	@0.05	@0.10	@0.15	@0.20
DINOv2	ViT-S	36.83	53.93	63.53	70.10
DINOv2	ViT-B	**38.43**	**55.36**	**64.73**	**70.99**
DINOv3	ViT-S	34.47	52.76	61.06	67.53
DINOv3	ViT-B	**37.56**	**54.06**	**62.44**	**67.91**
SAM	ViT-B	16.41	24.91	31.78	38.48

(b) PCK per point.

Table 1. Backbone comparison across PCK thresholds. We report both PCK per image and PCK per point. For DINOv2 and DINOv3 we evaluate ViT-S and ViT-B; for SAM only ViT-B is available. Best results within each model family are highlighted in bold.

Model	PCK@0.10 (image)	PCK@0.10 (point)
DINOv2 L0	52.46	55.26
DINOv2 L1	76.34	79.38
DINOv2 L2	**77.49**	80.67
DINOv2 L4	77.00	**80.84**

Table 2. Effect of light fine-tuning on DINOv2. L0 denotes the frozen backbone, while LX indicates fine-tuning of the last X transformer layers.

Model	PCK@0.10 (image)	PCK@0.10 (point)
DINOv3 L0	48.20	54.06
DINOv3 L1	74.90	78.02
DINOv3 L2	**76.80**	**79.85**
DINOv3 L4	75.06	78.71

Table 3. Effect of light fine-tuning on DINOv3. Best performance is achieved by fine-tuning a limited number of layers.

Model	PCK@0.10 (image)	PCK@0.10 (point)
SAM L0	21.53	24.91
SAM L1	41.47	45.61
SAM L2	42.88	47.64
SAM L4	**44.62**	**49.87**

Table 4. Effect of light fine-tuning on SAM. Unlike DINO-based models, performance consistently improves as more layers are fine-tuned.

stronger adaptation to effectively transfer to correspondence tasks.

4.7. Window Soft-Argmax

We evaluate the effect of replacing the standard argmax with a window soft-argmax (WSA), setting the window radius to $r = 3$ (window size 7×7) and temperature $\tau = 0.02$. As summarized in Tab. 6, WSA improves accuracy in the frozen setting, yielding consistent gains on DINOv2 and DINOv3, while improvements on SAM are smaller and can be mixed depending on the aggregation scheme. We attribute this variability to the noise present in off-the-shelf feature maps: while argmax ignores adjacent noisy pixels, WSA incorporates them into the weighted average, potentially causing a coordinate drift that limits precision. We further study whether WSA remains beneficial after task adaptation by applying it to the best fine-tuned variants of each backbone family (DINOv2-L2, DINOv3-L2, and SAM-L4). Tab. 5 shows that WSA provides consistent gains across all thresholds, indicating that smoother, locally constrained localization is complementary to fine-tuning. Unlike the frozen regime, fine-tuning suppresses background noise and produces sharper, more symmetric activation peaks. This allows WSA to effectively mitigate the quantization error of the integer argmax without suffering from noise-induced drift.

4.8. Cross-Dataset Robustness Evaluation

To evaluate cross-dataset robustness, we compare pre-trained and fine-tuned backbones performance on PF-Pascal, PF-Willow, and AP-10K without any additional adaptation on the target datasets (Tabs. 7 and 8). This comparative setting allows us to verify whether lightweight fine-tuning strategies effectively improve generalization under distribution shifts, given the distinct object categories and visual styles of the target datasets. The analysis of the results reveals a consistent trend in the behavior of the different backbones across the considered datasets. The main observations are as follows.

4.8.1. Performance on PF-Pascal compared to SPair-71k.

All backbones demonstrate higher performance on PF-Pascal than on SPair-71k. Although both datasets share the PASCAL VOC class set, ensuring high compatibility with the features learned during training, SPair-71k represents a more significant challenge. This is due to the intrinsic nature of SPair, which was specifically designed to include extreme variations in scale and viewpoint, as well as heavy occlusions, making it a much more rigorous benchmark com-

PCK	DINOv2-L2		DINOv3-L2		SAM-L4	
	Base	+WS	Base	+WS	Base	+WS
@0.05	55.35	**61.25**	54.98	**60.48**	27.41	**29.76**
@0.10	77.49	**80.03**	76.80	**79.17**	44.62	**47.10**
@0.15	85.02	**86.38**	84.36	**85.29**	54.63	**56.30**
@0.20	89.06	**89.69**	88.01	**88.63**	61.35	**62.38**

(a) PCK per image.

PCK	DINOv2-L2		DINOv3-L2		SAM-L4	
	Base	+WS	Base	+WS	Base	+WS
@0.05	58.66	**64.75**	58.38	**63.97**	30.98	**33.55**
@0.10	80.67	**83.20**	79.85	**82.25**	49.87	**52.65**
@0.15	87.70	**88.99**	86.88	**87.77**	60.32	**62.04**
@0.20	91.29	**92.89**	90.13	**90.67**	66.99	**68.04**

(b) PCK per point.

Table 5. Effect of window soft-argmax (WS) on fine-tuned models. "Base" denotes argmax, while "+WS" denotes window soft-argmax. We report results for the best fine-tuning configuration of each backbone family (DINOv2-L2, DINOv3-L2, SAM-L4).

PCK	DINOv2		DINOv3		SAM	
	Base	+WS	Base	+WS	Base	+WS
@0.05	35.88	**38.00**	**32.73**	32.21	**13.98**	12.69
@0.10	52.46	**54.55**	48.20	**49.70**	21.53	**21.55**
@0.15	61.82	**63.06**	56.51	**57.69**	27.78	**28.22**
@0.20	68.03	**68.90**	61.93	**62.94**	33.84	**33.94**

(a) PCK per image.

PCK	DINOv2		DINOv3		SAM	
	Base	+WS	Base	+WS	Base	+WS
@0.05	38.43	**40.69**	37.56	37.18	**16.41**	15.09
@0.10	55.36	**57.59**	54.06	**55.68**	24.91	**25.08**
@0.15	64.73	**65.92**	62.44	**63.73**	**32.78**	32.34
@0.20	70.99	**71.83**	67.91	**68.85**	38.18	**38.40**

(b) PCK per point.

Table 6. Effect of window soft-argmax (WS) on correspondence estimation. "Base" denotes hard argmax, while "+WS" denotes window soft-argmax. Results are reported for multiple PCK thresholds.

pared to the more linear PF-Pascal.

4.8.2. Limits of PF-Willow.

Despite its relatively simple structure, PF-Willow yields lower results than Pascal. Even though the categories are nominally similar, the different visual distribution and the limited size of the dataset make the features less effective at capturing the specific correspondences of this benchmark.

4.8.3. Challenge of AP-10K: From Pose Estimation to Generalization.

The AP-10K benchmark proves to be the most difficult. Being a dataset focused on Animal Pose Estimation, it requires anatomical precision that goes beyond simple semantic matching.

- **Cross-Species:** Complexity increases when mapping keypoints onto species never seen during training (though belonging to known families).
- **Cross-Family:** This represents the peak of complexity, requiring the model to abstract the concept of "anatomy" across animal families with completely different morphologies.

4.8.4. Impact of Fine-tuning.

Finally, a systematic and transversal improvement is observed for all architectures when comparing fine-tuned models with frozen ones. The ability to adapt the backbone weights to the specificities of the target domain allows the models to overcome the limitations of general-purpose representations, confirming the importance of task-specific adaptation for complex matching tasks.

5. Conclusion

This study demonstrates that visual foundation models are highly effective for semantic correspondence when paired with the right adaptation strategies. Our key findings show that lightweight fine-tuning of just 1–2 layers provides a massive performance boost over frozen features. Additionally, the window soft-argmax strategy effectively improves spatial precision. While models like DINOv2 and DINOv3 exhibit strong generalization across datasets like PF-Pascal and AP-10K, significant taxonomic variations still represent a boundary for current zero-shot and fine-tuned representations.

	SPair-71k			AP-10K Test			AP-10K C.S.			AP-10K C.F.			PF-Pascal			PF-Willow		
Backbone	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15
DINOv2 L0	40.69	57.59	65.92	43.74	64.05	75.11	39.96	59.99	72.36	32.02	49.77	62.44	64.66	83.51	90.35	38.29	65.87	82.27
DINOv2 L2	**64.75**	**83.20**	**88.98**	**50.93**	**69.59**	**78.95**	**46.71**	**64.31**	**74.27**	**38.89**	**56.83**	**66.99**	70.72	87.24	93.04	52.53	76.60	87.14
DINOv3 L0	37.18	55.68	63.72	44.72	62.49	72.23	36.53	56.57	67.77	28.29	44.54	55.40	67.98	86.95	**93.29**	42.98	72.18	84.52
DINOv3 L2	63.97	82.25	88.77	48.16	66.90	76.87	44.40	62.95	72.93	35.74	54.30	64.63	**72.96**	**87.74**	92.33	**52.70**	**77.56**	**88.06**
SAM L0	15.09	25.08	32.34	14.28	26.22	36.45	10.20	20.89	30.61	6.05	14.04	22.37	33.72	47.97	57.17	22.28	39.88	49.84
SAM L4	33.55	52.65	62.04	24.60	42.23	53.41	19.92	37.55	49.31	14.27	29.17	40.08	42.38	59.73	67.85	34.59	54.35	64.13

Table 7. PCK per point matching performance on SPair-7k, AP-10K (Test, Cross-Species and Cross-Family), PF-Pascal and PF-Willow.

	SPair-71k			AP-10K Test			AP-10K C.S.			AP-10K C.F.			PF-Pascal			PF-Willow		
Backbone	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15
DINOv2 L0	38.00	54.54	63.05	46.76	65.18	75.58	42.20	60.92	71.15	35.89	52.85	64.28	62.96	82.15	89.23	38.28	65.86	81.26
DINOv2 L2	**61.24**	**80.02**	**86.37**	**54.87**	**71.23**	**79.53**	**50.79**	**66.19**	**74.94**	**43.55**	**60.55**	**69.49**	70.31	86.60	92.50	52.53	76.6	87.14
DINOv3 L0	32.21	49.70	57.69	44.69	63.58	73.64	39.36	58.53	69.20	30.98	47.79	57.80	66.90	86.24	**92.60**	41.97	72.17	84.52
DINOv3 L2	60.47	79.16	85.28	52.49	68.91	77.70	48.39	64.33	72.97	42.37	58.92	67.54	**71.30**	**86.95**	91.48	**52.70**	**77.56**	**88.05**
SAM L0	12.69	21.54	28.21	15.06	26.38	36.33	10.73	20.98	30.57	6.54	14.26	22.63	32.81	46.92	56.41	22.27	39.87	49.84
SAM L4	29.75	47.10	56.29	26.13	43.02	53.69	20.99	38.11	49.57	15.04	30.20	41.08	41.46	58.73	66.91	34.58	54.35	64.13

Table 8. PCK per image matching performance on SPair-7k, AP-10K (Test, Cross-Species and Cross-Family), PF-Pascal and PF-Willow

References

[1] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers, 2021. 2

[2] Bumsub Ham, Minsu Cho, Cordelia Schmid, and Jean Ponce. Proposal flow: Semantic correspondences from object proposals, 2017. 1, 4

[3] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. *arXiv:2304.02643*, 2023. 1, 2, 5

[4] Xinghui Li, Jingyi Lu, Kai Han, and Victor Prisacariu. Sd4match: Learning to prompt stable diffusion model for semantic matching, 2024. 3

[5] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019. 5

[6] Juhong Min, Jongmin Lee, Jean Ponce, and Minsu Cho. Spair-71k: A large-scale benchmark for semantic correspondence, 2019. 1, 4, 5

[7] Maxime Oquab, Timothée Darcet, Theo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Russell Howes, Po-Yao Huang, Hu Xu, Vasu Sharma, Shang-Wen Li, Wojciech Galuba, Mike Rabbat, Mido Assran, Nicolas Ballas, Gabriel Synnaeve, Ishan Misra, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision, 2023. 1, 5

[8] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. Dinov3, 2025. 1, 5

[9] Luming Tang, Menglin Jia, Qianqian Wang, Cheng Perng Phoo, and Bharath Hariharan. Emergent correspondence from image diffusion, 2023. 1, 2, 4

[10] Hang Yu, Yufei Xu, Jing Zhang, Wei Zhao, Ziyu Guan, and Dacheng Tao. Ap-10k: A benchmark for animal pose estimation in the wild, 2021. 1, 4

[11] Junyi Zhang, Charles Herrmann, Junhwa Hur, Luisa Polania Cabrera, Varun Jampani, Deqing Sun, and Ming-Hsuan Yang. A tale of two features: Stable diffusion complements dino for zero-shot semantic correspondence, 2023. 2

[12] Junyi Zhang, Charles Herrmann, Junhwa Hur, Eric Chen, Varun Jampani, Deqing Sun, and Ming-Hsuan Yang. Telling left from right: Identifying geometry-aware semantic correspondence, 2024. 1, 2, 3